Практическая работа.
"Обратная матрица"
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1. Найти матрицу обратную данной.

$$A=\begin{pmatrix}1 & -2 & 0\\ 3 & 5 & -7\\ -4 & 1 & 2\end{pmatrix} \qquad B=\begin{pmatrix}1 & -2\\ -3 & 4\end{pmatrix}$$

Формула: $A^{-1}=\frac{1}{\det A}\cdot\tilde{A}^{T}$

$$\tilde{A}=\begin{pmatrix}A_{11} & A_{12} & A_{13}\\ A_{21} & A_{22} & A_{23}\\ A_{31} & A_{32} & A_{33}\end{pmatrix}$$

Сделать проверку: $A\cdot A^{-1}=E$

$B\cdot B^{-1}=E$

Найдем A^{-1}

$$\det A=\begin{vmatrix}1 & -2 & 0\\ 3 & 5 & -7\\ -4 & 1 & 2\end{vmatrix}=1\cdot5\cdot2+0\cdot3\cdot1+(-4)(-2)(-7)-0\cdot5\cdot(-4)-$$
$$-2(-2)\cdot3-1\cdot1\cdot(-7)=10-56+12+7=-27$$

$\det A\neq0 \Rightarrow$ обратная матрица существует

$$A_{11}=(-1)^{1+1}\cdot M_{11}=1\cdot\begin{vmatrix}5 & -7\\ 1 & 2\end{vmatrix}=5\cdot2-(-7)\cdot1=10+7=17$$

$$A_{12}=(-1)^{1+2}\cdot M_{12}=(-1)\cdot\begin{vmatrix}3 & -7\\ -4 & 2\end{vmatrix}=(-1)\left(3\cdot2-(-7)\cdot(-4)\right)=(-1)\cdot(6-28)=22$$

$$A_{13}=(-1)^{1+3}\cdot M_{13}=1\cdot\begin{vmatrix}3 & 5\\ -4 & 1\end{vmatrix}=3\cdot1-5\cdot(-4)=3+20=23$$

$$A_{21}=(-1)^{2+1}\cdot M_{21}=(-1)\cdot\begin{vmatrix}-2 & 0\\ 1 & 2\end{vmatrix}=(-1)\cdot\left((-2)2-0\cdot1\right)=(-1)\cdot(-4)=4$$

$$A_{22}=(-1)^{2+2}\cdot M_{22}=1\cdot\begin{vmatrix}1 & 0\\ -4 & 2\end{vmatrix}=1\cdot2-0\cdot(-4)=2$$

$$A_{23}=(-1)^{2+3}\cdot M_{23}=(-1)\cdot\begin{vmatrix}1 & -2\\ -4 & 1\end{vmatrix}=(-1)\cdot\left(1\cdot1-(-2)(-4)\right)=(-1)(1-8)=7$$

$$A_{31}=(-1)^{3+1}\cdot M_{31}=1\cdot\begin{vmatrix}-2 & 0\\ 5 & -7\end{vmatrix}=(-2)(-7)-0\cdot5=14$$

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РУП 11-26 03

$$A_{32} = (-1)^{3+2} \cdot M_{32} = (-1) \cdot \begin{vmatrix} 1 & 0 \\ 3 & -7 \end{vmatrix} = (-1) \cdot (1 \cdot (-7) - 0 \cdot 3) = (-1)(-7) = 7$$

$$A_{33} = (-1)^{3+3} \cdot M_{33} = 1 \cdot \begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix} = 1 \cdot 5 - (-2) \cdot 3 = 5 + 6 = 11$$

$$\tilde{A} = \begin{pmatrix} 17 & 22 & 23 \\ 4 & 2 & 7 \\ 14 & 7 & 11 \end{pmatrix} \quad \tilde{A}^T = \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix}$$

$$A^{-1} = \frac{1}{-27} \cdot \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix} = -\frac{1}{27} \cdot \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix}$$

Проверка:

$$A \cdot A^{-1} = -\frac{1}{27} \cdot \begin{pmatrix} 1 & -2 & 0 \\ 3 & 5 & -7 \\ -4 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 17 & 4 & 14 \\ 22 & 2 & 7 \\ 23 & 7 & 11 \end{pmatrix} = -\frac{1}{27} \cdot \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} =$$

$$= -\frac{1}{27} \cdot \begin{pmatrix} -27 & 0 & 0 \\ 0 & -27 & 0 \\ 0 & 0 & -27 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = E$$

$c_{11} = 1 \cdot 17 + (-2) \cdot 22 + 0 \cdot 23 = 17 - 44 = -27$

$c_{12} = 1 \cdot 4 + (-2) \cdot 2 + 0 \cdot 7 = 4 - 4 = 0$

$c_{13} = 1 \cdot 14 + (-2) \cdot 7 + 0 \cdot 11 = 14 - 14 = 0$

$c_{21} = 3 \cdot 17 + 5 \cdot 22 + (-7) \cdot 23 = 51 + 110 - 161 = 0$

$c_{22} = 3 \cdot 4 + 5 \cdot 2 + (-7) \cdot 7 = 12 + 10 - 49 = -27$

$c_{23} = 3 \cdot 14 + 5 \cdot 7 + (-7) \cdot 11 = 42 + 35 - 77 = 0$

$c_{31} = (-4) \cdot 17 + 1 \cdot 22 + 2 \cdot 23 = -68 + 22 + 46 = 0$

$c_{32} = (-4) \cdot 4 + 1 \cdot 2 + 2 \cdot 7 = -16 + 2 + 14 = 0$

$c_{33} = (-4) \cdot 14 + 1 \cdot 7 + 2 \cdot 11 = -56 + 7 + 22 = -27$

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Найдем B^{-1}

$$\det B = \begin{vmatrix} 1 & -2 \\ -3 & 4 \end{vmatrix} = 1 \cdot 4 - (-2)(-3) = 4 - 6 = -2$$

$\det B \neq 0 \Rightarrow$ обратная матрица существует

$$B_{11} = (-1)^{1+1} \cdot M_{11} = 1 \cdot 4 = 4$$

$$B_{12} = (-1)^{1+2} M_{12} = (-1) \cdot (-3) = 3$$

$$B_{21} = (-1)^{2+1} M_{21} = (-1)(-2) = 2$$

$$B_{22} = (-1)^{2+2} M_{22} = 1 \cdot 1 = 1$$

$$\tilde{B} = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \qquad \tilde{B}^T = \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}$$

$$B^{-1} = \frac{1}{-2} \cdot \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}$$

Проверка:

$$B \cdot B^{-1} = -\frac{1}{2} \cdot \begin{pmatrix} 1 & -2 \\ -3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} = -\frac{1}{2} \cdot \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} =$$

$$= -\frac{1}{2} \cdot \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = E$$

$$c_{11} = 1 \cdot 4 + (-2) \cdot 3 = 4 - 6 = -2$$

$$c_{12} = 1 \cdot 2 + (-2) \cdot 1 = 2 - 2 = 0$$

$$c_{21} = (-3) \cdot 4 + 4 \cdot 3 = -12 + 12 = 0$$

$$c_{22} = (-3) \cdot 2 + 4 \cdot 1 = -6 + 4 = -2$$